

Characterization-Guided Framework for Structural Graph Augmentation for GNNs

Muhammad Safi Ullah ADAM

Data and Intelligence for Smart Systems (DISS)

Supervisors

Hamida SEBA – LIRIS

Angela BONIFATI – Lyon1 Tutor

Date: 28th August, 2026

Outline

Introduction

Related Work

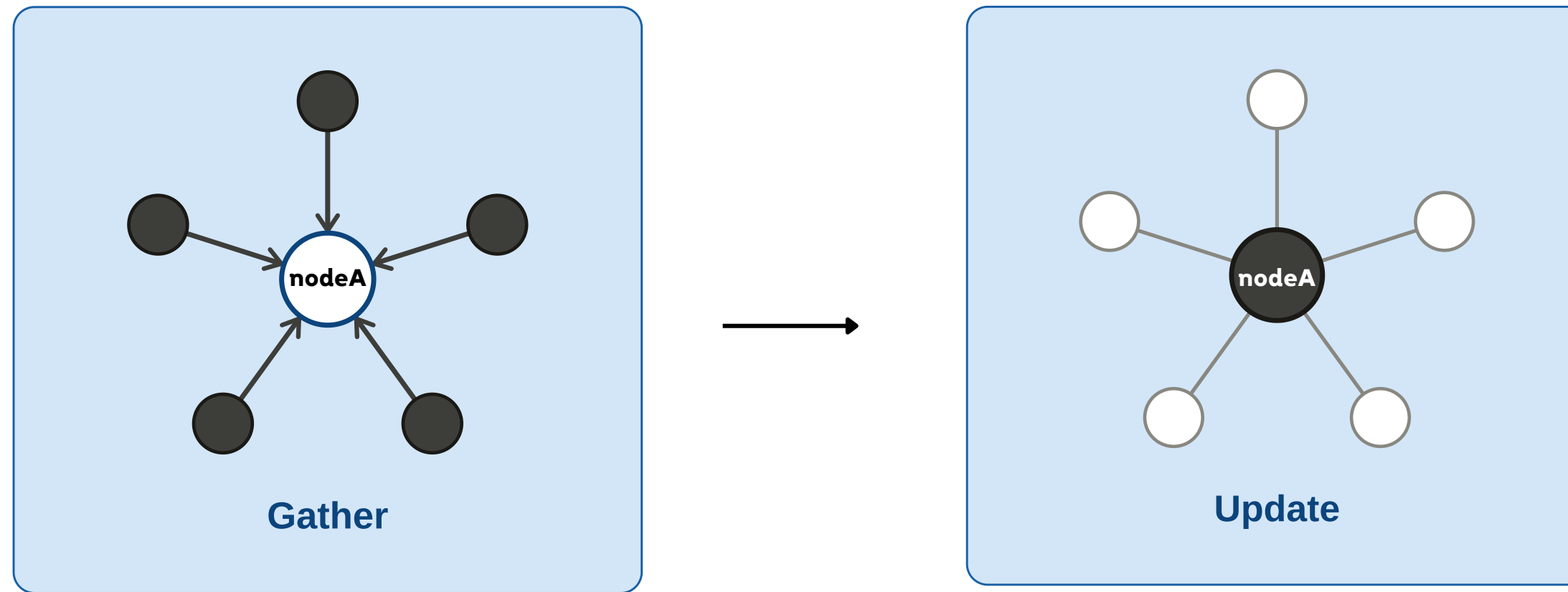
Proposed Solution

Research Methodology

Results and Discussion

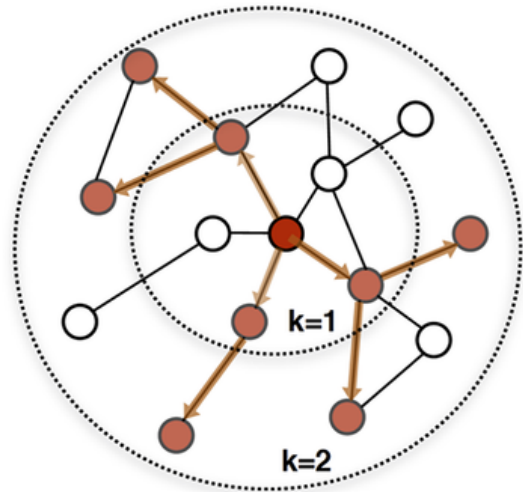
Conclusion and Future Work

Introduction

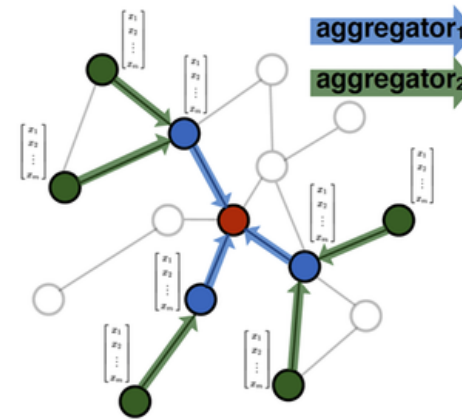


GNNs learn by aggregating information from connected neighbours.

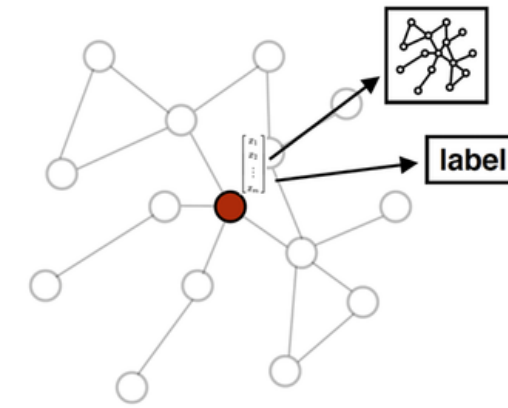
Problem Statement



1- Sample Neighborhood



2- Aggregate feature information
from neighbors



3- Predict graph context and labels
using aggregated information

*image taken from original GraphSAGE paper (Hamilton et al., 2017)

GNNs only exchange information along existing edges

Physical connection $\neq$ structural similarity

This can limit message passing when structurally similar nodes are physically distant.

Related Work

Structural Identity Learning

Learn node representations from structural roles, allowing distant nodes with similar graph functions to be represented similarly.

Heterophilic GNN Architectures

Redesign model architecture to better handle graphs where connected nodes often belong to different classes. (Model-level fix)

Graph Characterization

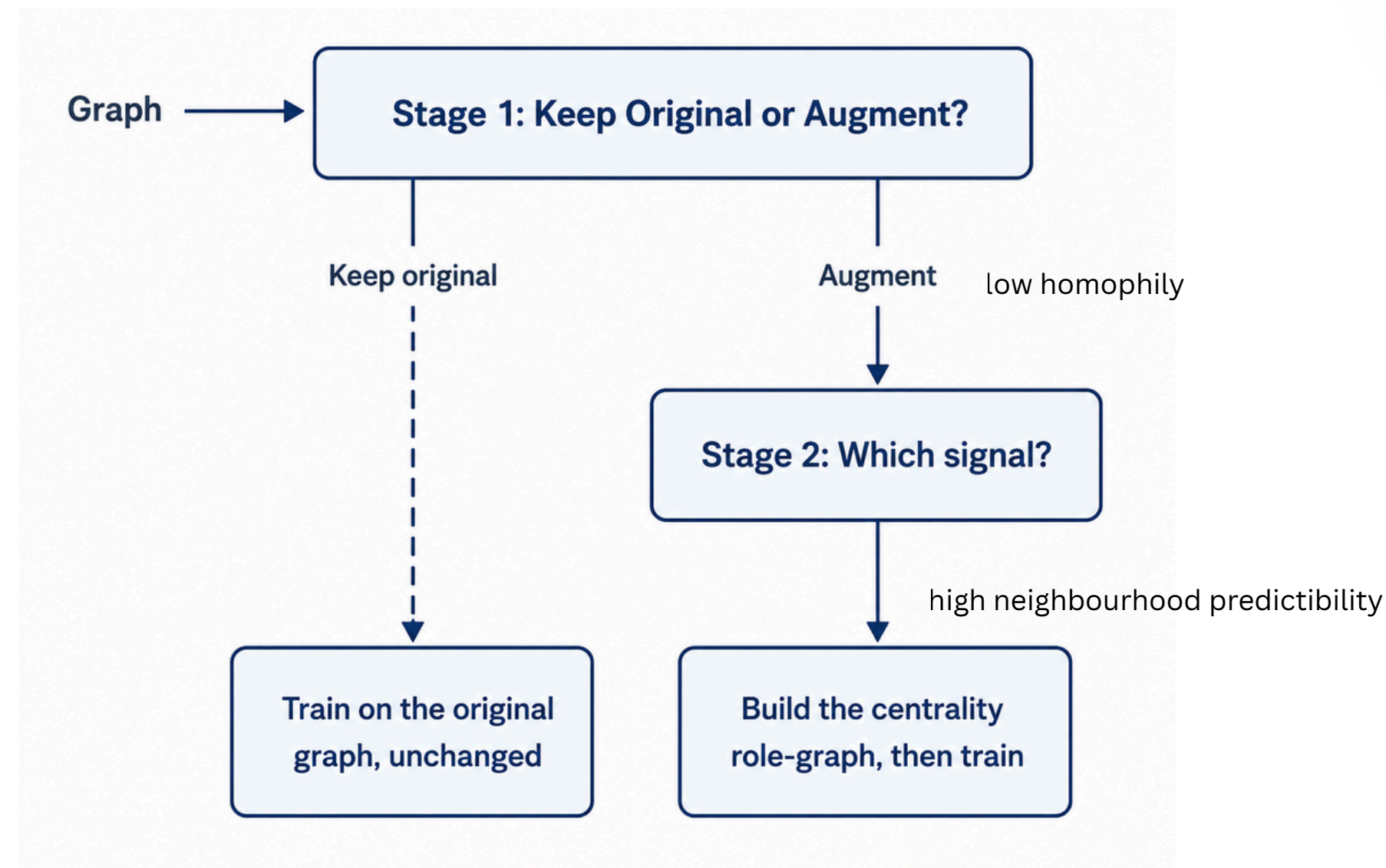
Systematically evaluate homophily measures to describe how graph structure and labels interact across datasets.

Measure graph → understand graph behaviour

Automatic Strategy Selection

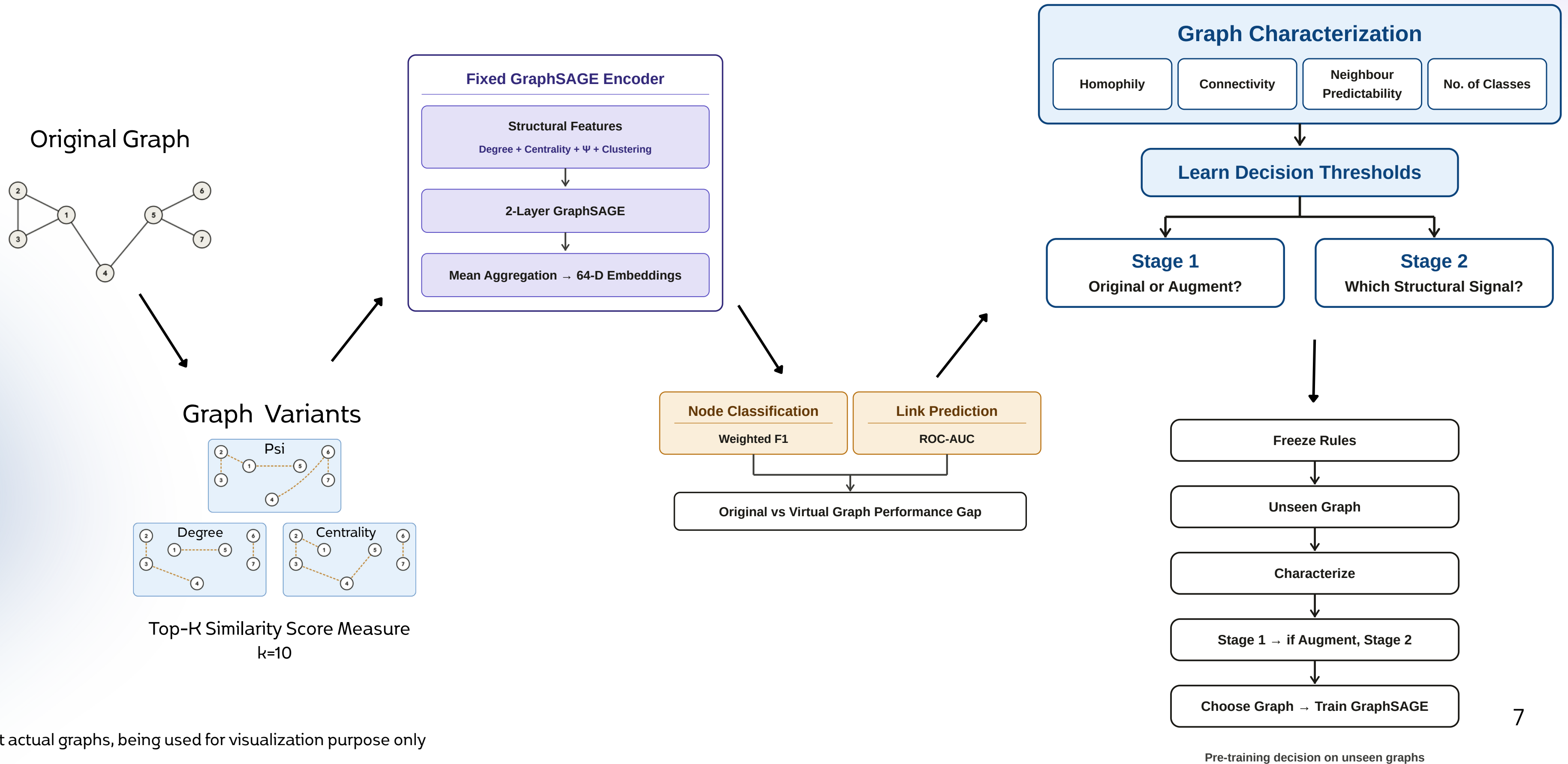
Automatically search for suitable augmentation strategies searching and optimizing many possibilities during model development using AutoML/RL and Topology/GNN Architecture.

Proposed Solution

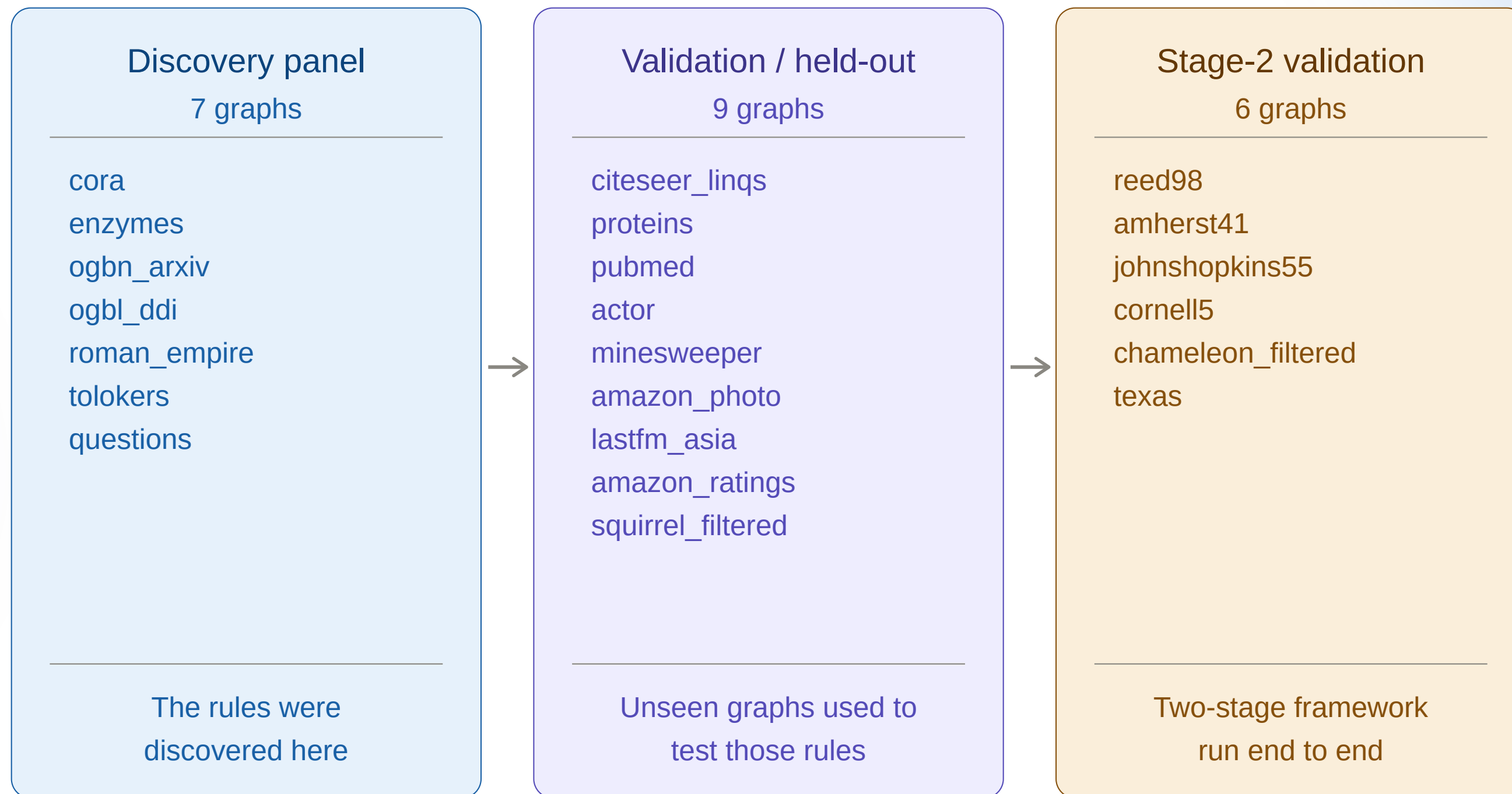


- **Structural augmentation:** Measure 7 graph characteristics (homophily, neighbourhood predictability, connectivity, clustering, number of classes)
- Graph Characteristics -> Learn the performance gap (original - augmented).
- Characterize first, then **augment before GNN training.**

Methodology



Methodology



All the datasets we used are publicly available: <https://github.com/SafiUllahAdam/Structure-Aware-Graph-Augmentation-for-GNNs>

Results

Candidate	Decision Rule	Fit Interval	Spearman ρ	Final Status
Adjusted Homophily - R1	Augment if < 0.227	0.093–0.361	–0.900 / –1.000	Final Stage-1 Rule
Largest Component Fraction - R2	Augment if $> \mathbf{0.959}$	0.918–1.000	+0.780 / +0.780	considered
Neighbour Predictability	Augment if $< \mathbf{0.408}$	0.332–0.484	–0.700 / –0.600	Rejected

Candidate Stage-1 Rule

low adjusted homophily $\rightarrow$ consider structural augmentation

7/9 correct predictions, Strong correlation with augmentation

Results

10

Results 2 - Stage-1 Rule Validation

Dataset	Adj. Hom.	LCC	R1 Prediction	R2 Prediction	Actual LP Verdict	Original	Best Aug.
citeseer_linqs	0.6731	0.6464	Keep original	Keep original	Keep original	0.6218	0.5437
proteins	0.3552	0.0143	Keep original	Keep original	Keep original	0.6721	0.5834
pubmed	0.6861	1	Keep original	Augment	Keep original	0.6392	0.6238
actor	0.0028	1	Augment	Augment	Augment	0.5953	0.6617
minesweeper	0.0094	1	Augment	Augment	Keep original	0.7093	0.6671
amazon_photo	0.7851	0.9787	Keep original	Augment	Tie	0.7857	0.7803
lastfm_asia	0.8562	1	Keep original	Augment	Keep original	0.7218	0.7002
amazon_ratings	0.1402	1	Augment	Augment	Keep original	0.7536	0.6551
squirrel_filtered	0.0086	1	Augment	Augment	Augment	0.7399	0.7736

Results

Results 3 - Stage-2 Signal Selection

Dataset	Adjusted Neighbour Predictability	Observed Winner(s)	Stage-2 Prediction
tolokers	-0.3274	Ψ / degree	Ψ or degree, unresolved
questions	-0.0041	degree	Ψ or degree, unresolved
actor	0.006	degree	Ψ or degree, unresolved
squirrel_filtered	0.0123	centrality / degree	centrality
roman_empire	0.3324	centrality	centrality
pubmed	0.6946	centrality	centrality
lastfm_asia	0.8498	centrality	centrality

Low neighbour predictability $\rightarrow \Psi$ / Degree region

Higher neighbour predictability $\rightarrow$ Centrality region

Results

Results 3 - Stage-2 Signal Validation

Dataset	Stage 1 - keep/augment	Stage 2 Used?	Stage-2 Prediction	Centrality Score	Original Graph	Interpretation
Chameleon- filtered	KEEP	No	N/A	–	–	Stopped at Stage 1
Texas	KEEP	No	N/A	–	–	Stopped at Stage 1
Reed98	KEEP	No	N/A	–	–	Stopped at Stage 1
Amherst41	AUGMENT	Yes	Centrality	0.675	0.665	Marginally better
JohnsHopkins 55	AUGMENT	Yes	Centrality	0.716	0.685	Centrality better
Cornell5	AUGMENT	Yes	Centrality	0.708	0.691	Centrality better

Conclusion

- no **one-size-fits-all** graph topology: the best choice depends on the dataset and downstream task.
- Adjusted homophily provides a useful Stage-1 rule.
- Stage 2 showed a promising Centrality-selection pattern.

Limitations & Future Work

- **Adaptive Top-K selection:** Select top-K based on graph data
- **Label Dependant Rules**
- **Stage 2 selection-patterns** for Ψ similarity / degree and more.
- **Adaptive Generalization:** adapt for other tasks like anomaly detection.
- **Extend and validate the framework:** test with other GNNs (GIN/GAT)